MIL-D-46539 (Ord)
6 June 1961

MILITARY SPECIFICATION

DESTRUCTOR, EXPLOSIVE, M19
LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of initiator designated as Destructor, Explosive, M19.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieves; Standard, Testing.

MILITARY

MIL-G-2550 - General Specification for Ammunition Except Small Arms Ammunition.
MIL-I-45208 - Inspection Requirements, General Specification for.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definitions.

FSC 1375

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DRAWINGS

ORDNANCE CORPS

8846587 - Destructor, Explosive, M19, Assembly.
8796674 - Ammunition Container, Fiber, M- ., for Destructor, Explosive, M19, Assembly.
8796675 - Box, Packing, Ammunition for Destructor, Explosive, M19, Assembly, in Fiber Container M- .

PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-78 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations (1949 Edition and revisions) available from the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders for the above publication should cite "49 CFR 71-78 (latest revision)".

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3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Assemblies.-The assemblies shall comply with all requirements specified on Drawing (dwg.) 8846587 and with all requirements specified in applicable specifications.

3.3 Composition of booster pellet.

3.3.1 Certification.-The contractor shall certify that the molding powder contains the specified percentages and classes of polystyrene, dioctylphthalate and RDX.

3.3.2 Foreign matter.-The molding powder shall be free of metallic inclusions and wooden particles when examined as specified in 4.3.2.

3.3.3 Moisture content.-The molding powder shall not contain more than 0.3 percent maximum (max.) when determined as specified in 4.3.5.

3.3.4 Granulation.-The granulation of the molding powder shall be as indicated below when determined as specified in 4.3.6.

U.S. Standard Sieve No.	Percent, Minimum (min.)
18	through 100
30	through 95
60	retained 90
80	retained 98
100	retained 99.9

3.4 Granulation of magnesium.-The contractor shall certify that the magnesium in the molding powder for the magnatol pellets is of the granulation, by percentages, specified on the drawing.

3.5 Workmanship.-All parts and assemblies shall be clean and free from surface defects, chips, dirt, grease, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Surface coatings shall be continuous except for a few small scratches. All required marking shall be neat and sharply defined.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. The provisions of Specification MIL-G-2550 shall apply.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system in accordance with Specification MIL-I-45208.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisons of the approved current written description of the system.

b. Number of units of product inspected.

c. Results obtained for all inspection performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of analyses on all material procured directly by the contractor when such material is controlled by Government specifications listed in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL-I-45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 Lot formation.-A lot shall consist of loaded initiators produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance, or fit, as determined by the Government, shall not necessitate changing the lot number. Each lot shall contain:

a. Metal parts from one manufacturer bearing the same interfix number.

b. Pellets from one batch or homogeneous blend of magnatol.

c. Booster pellets from one batch or homogenous blend of batches.

4.3 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall 100 percent. Continuous sampling plans in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

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4.2.2.1 Assembly, minus end plug (see dwg. 8846587).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.25 percent	
101.	Foreign matter in cavity	Visual
Minor:	None defined	

4.2.2.2 Loaded assembly, unclosed (see dwg. 8846587).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.25 percent	
101.	Explosive weight, minimum (min.)...	Balance
102.	Booster pellet improperly positioned	Visual-Manual
Minor:	AQL 1.00 percent	
201.	Pad missing	Visual
202.	Evidence of poor workmanship (see 3.5)	Visual

4.2.2.3 Assembly, loaded (see dwg. 8846587).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.25 percent	
101.	Ogive not removable	Manual
Minor:	AQL 2.50 percent	
201.	Shipping plug and washer not secure	Manual
202.	End plug not staked	Visual
203.	Paint missing or inadequate	Visual
204.	Marking missing, misleading or unidentifiable	Visual
205.	Evidence of poor workmanship (see 3.5) .	Visual

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4.2.2.4 Fiber container, unsealed (see dwg. 8796674).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	None defined	
Minor:	AQL 1.50 percent	
201.	Destructor cannot be removed by hand ...	Manual
202.	Packing component missing	Visual
203.	Marking missing, misleading or unidentifiable	Visual

4.2.2.5 Fiber container, sealed (see dwg. 8796674).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.65 percent	
101.	Tape incomplete or badly wrinkled	Visual
102.	Metal end loose or distorted	Visual-Manual
103.	Cut, scuffs, gouges or checks which penetrate outer layers	Visual
Minor:	AQL 2.50 percent	
201.	Length of tear tab, min.	Gage
202.	Gap between body and cover in excess of 1/8 inch.....	Gage
203.	Marking misleading or unidentifiable ...	Visual
204.	Contents loose	Manual

4.2.2.6 Box, packing, unsealed see dwg. 8796675.

Categories	Defects	Method of inspection
Critical	None defined	
Major	AQL 0.25 percent	
101.	Quantity incorrect	Visual
102.	All sealing strips at one end	Visual-Manual
Minor	AQL 1.00 percent	
201.	Foreign matter inside	Visual
202.	Filler missing	Visual

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4.2.2.7 Box, packing, sealed (see dwg. 8796675).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.65 percent	
101.	Hardware missing broken or loose	Visual-Manual
102.	Strapping missing, broken or loose	Visual-Manual
103.	Handle missing or insecure	Visual-Manual
104.	Contents exposed	Visual
105.	Board broken, split or curled	Visual
106.	DOD symbol misleading or unidentifiable	Visual
Minor:	AQL 2.50 percent	
201.	Hardware improperly engaged	Visual-Manual
202.	Contents loose (when box is tilted).....	Manual
203.	Strapping improperly located	Visual
204.	Marking missing, misleading or unidentifiable	Visual
205.	Car seal missing, unsealed or improperly positioned	Visual

4.2.3 Testing.

4.2.3.1 Specific gravity of disc, booster and ring pellets (see dwgs. 8846592, 8846593 and 8846594, respectively) - Major defects.-A sample of 10 each of each pellet shall be selected from each applicable lot for this test. If two or more pellets fail to meet their applicable requirement, the lot represented shall be rejected. If one pellet fails, a second sample of 20 pellets shall be selected from the represented lot and subjected to test. The lot shall be rejected and testing may be discontinued if one pellet fails to meet the applicable requirement during the testing of the second sample.

4.2.3.2 Composition of booster pellets (see dwg. 8846587 3.3.2, 3.3.3 and 3.3.4) - Major defect.-Five samples of PBX of 8 ounces each, each from a different batch and representative of the batch, shall be selected for these determinations. If the number of batches per lot is less than five, all batches shall be sampled. The batch samples shall be thoroughly blended into a

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sample representing the lot. After examination for compliance with 3.3.2, the sample shall be tested for compliance with 3.3.3 and 3.3.4 and for RDX content. (see 6.2) Failure to meet one or more of the requirements shall reject the lot. If rejection occurs, the contractor may have each batch tested separately. Those batches meeting all requirements shall be considered acceptable.

4.2.3.3 Composition of magnatol (see dwg. 8846587) - Major defect.-A representative sample of 5 ounces shall be selected from each batch in the lot and shall then be thoroughly blended. Failure to meet the requirements of the drawing when tested as specified in 4.3.3 shall reject the lot. If rejection occurs, the contractor may have each batch tested separately. Those batches meeting the requirements shall be considered acceptable.

4.2.4 Inspection equipment.-For the performance of the examinations and tests specified in 4.2.2 and 4.2.3, when applicable, the contractor shall have available, and utilize correctly, all necessary gages, measuring, and testing equipment (see 6.3). When inspecting the contractor's inspection equipment, the Government inspector will determine that the contractor has available, and utilizes correctly, gages, measuring and testing equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. The contractor will have available a set of master gages, standards, and appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration will be maintained by the contractor and made available for review by the Government. The calibration of gages, standards, and instruments will be periodically checked by authorized Government personnel.

4.3 Test methods and procedures.

4.3.1 Specific gravity.-The pellet shall be weighed on a balance and the weight recorded as dry weight. It shall then be weighed completely submerged under water, the temperature of which shall be 68 degrees Fahrenheit (F.) plus or minus 2 degrees F., and the weight recorded. Specific gravity shall be calculated by dividing the dry weight by the difference between the dry weight and the weight in water.

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4.3.2 Foreign matter.-The sample shall be visually examined for the presence of any material or matter not specified in the composition by pouring the sample slowly from one container to another several times. Presence of foreign matter shall class the lot defective.

4.3.3 Composition of magnatol.-Transfer a 10 gram portion of the sample to a tared medium porosity filtering crucible and dissolve the TNT by washing with small portions of warm benzene. Evacuate the liquid from the crucible by vacuum. After all TNT has been dissolved, apply the vacuum until no odor of benzene is detected in the crucible. The crucible and residue shall then be dried at 100 degrees Centigrade plus or minus 2 degrees Centigrade for one half hour, cooled in a desiccator and weighed. Calculate the increase in weight of the crucible as percentage of magnesium by dividing the weight of the crucible after drying minus the original crucible weight times 100 by the original sample weight. The TNT content shall be determined by deducting the percent magnesium content from 100.

4.3.4 Determination of RDX content.-The determination shall be performed as follows: Weigh together a 50 milliliter Erlenmeyer stoppered flask, minus the stopper, and a 30 milliliter fritted glass crucible of medium porosity and record the weight. Weigh into the tared flask an accurately weighed portion of approximately 5 grams of the sample. Add 20 milliliters of toluene saturated with RDX. Stopper the flask and shake on a wrist action shaker for 20 minutes. Place the tared fritted glass crucible on a vacuum filtrator assembly and apply vacuum. Transfer as much as possible of the contents of the flask to the crucible, using five 15 milliliter portions of toluene saturated with RDX. Wash the stopper with the RDX saturated toluene solution and pass the washings through the crucible. Pour 10 milliliters of ether into the flask, swirl, and empty into the filtering crucible. Air dry the residue in the flask and crucible until no odor of ether can be detected. Dry the flask and crucible in an oven maintained at 65 degrees Centigrade plus or minus 2 degrees Centigrade for one hour. Weigh the flask and crucible together and record the increase in weight as residue. Calculate the percent RDX by dividing the weight of the residue times 100 by the weight of the original sample. (see 6.2). In case of dispute or failure, report the average of three determinations.

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4.3.5 Determination of moisture content.-The determination shall be performed as follows: transfer an accurately weighed portion of approximately 5 grams of the sample into a previously dried and weighed aluminum dish (approximately 60 millimeters in diameter and 30 millimeters deep). Place the sample in a vacuum oven pressure not greater than 8 centimeters of mercury) and maintain a temperature of 55 degrees Centigrade plus or minus 2 degrees Centigrade for 4 hours. Remove from the oven, weigh and determine the loss in weight. Calculate the moisture content by dividing the loss in weight of the sample times 100 by the original weight of the sample (see 6.2).

4.3.6 Granulation.-A 100 gram plus 0.1 gram portion of the sample shall be placed on the specified sieve or nest of sieves, complying with Specification RR-S-366, properly superimposed and assembled with a bottom pan. The nest of sieves, shall be covered and shaken for 3 minutes by means of a mechanically operated sieve shaker which shall impart to the sieves a rotary motion and a tapping action of 285 to 315 gyrations and 140 to 160 taps of the striker per minute. The portions passed or retained by the various sieves shall be weighed and the results calculated to a cumulative percentage basis.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Level A.-The destructors shall be packaged in accordance with dwg. 8796674.

5.2 Packing and marking.

5.2.1 Level A.-The packaged assemblies shall be packed and marked in accordance with dwg. 8796675.

5.3 Data cards.-Data card information shall be as specified in Specification MIL-G-2550.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

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6.2 In rounding off, if the digit to be discarded is 0 to 4 the last effective digit shall remain the same; if the digit to be discarded is 5 to 9 the last effective digit shall be increased by 1.

6.3 Inspection equipment.-Inspection equipment designs were not available at the time this specification was prepared. Prior to procurement, the procuring agency should contact Picatinny Arsenal, Dover, New Jersey, ATTN: ORDBB-ND, to determine availability of inspection equipment designs.

Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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